

César Díaz Blanco

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SUMMARY

3D Computer Vision specialist with expertise in multi-view reconstruction, generative models, and feed-forward architectures. Proven track record of designing scalable pipelines, modernizing cloud CI/CD dependency management, and reducing cloud storage costs. Recent research focuses on geometrically accurate Feed-Forward Gaussian Splatting.

EDUCATION

Master of Science in Machine Learning | *University of Tübingen* 2023–26

- ▶ GPA: 1.61/1.00
- ▶ Relevant Coursework: Virtual Humans, 3D Vision, Massive Parallel Computing, Computer Vision Seminar

Bachelor of Science in Engineering Physics | *University of Illinois at Urbana-Champaign (UIUC)* 2019–23

- ▶ Graduated with Honors (GPA: 3.94/4.00) · Computer Science Minor (GPA: 4.00/4.00)
- ▶ Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Systems Programming, Deep Learning for Computer Vision, Computational Photography, and Machine Perception

EXPERIENCE

Master Thesis  | *Autonomous Vision Group* , *Tübingen AI Center* September 2025–May 2026

Topic: Geometrically Accurate Feed-Forward Gaussian Splatting for Unbounded Scenes

Advisors: Zehao Yu, Haofei Xu · **Examiners:** Prof. Dr.-Ing. Andreas Geiger, Prof. Dr. Gerard Pons-Moll

- ▶ Implemented a 3D Smoothing Filter to limit the frequency of predicted Gaussians, thus stabilizing training and enabling encoder fine-tuning on new renderers and self-supervised geometric losses
- ▶ Improved mesh reconstruction precision on the Tanks and Temples (TnT) dataset by 73% over the photometric-only encoder, matching per-scene optimization performance
- ▶ Reduced absolute relative error for depth prediction across MipNeRF360, TnT, and DL3DV by 7% against the photometric-only encoder and 31% against Depth Anything 3

Research Project  | *Real Virtual Humans Group* , *Tübingen AI Center* January 2025–October 2025

Topic: Text-Driven Multi-View Fashion Editing

Advisors: Dr. Garvita Tiwari, Berna Kabadayi, Nikita Kister · **Examiners:** Prof. Dr. Gerard Pons-Moll

- ▶ Designed a novel two-stage, model-agnostic framework for text-driven, multi-view consistent editing
- ▶ Developed an evaluation suite for the outer garment removal task with three key metrics: 3D consistency of the generated clothing, VLM-judged generation quality and removal success, and preservation of unmodified regions
- ▶ Achieved a 90% removal success rate on the 4D-DRESS dataset and validated 3D consistency by optimizing a 3DGS representation on 10 generated views, yielding 28.44 PSNR on 10 held-out generated images

Working Student in Computer Vision and Software Engineering | *DeepScenario*  September 2024–May 2025

- ▶ Designed a pipeline for multi-view detection and tracking from ego camera sensors
- ▶ Prepared the data release of reconstructed scenes by developing mesh rendering scripts for thumbnails and mesh cleaning scripts to reduce mesh extent according to the tracked vehicles
- ▶ Modernized Python dependency management by migrating to uv, ensuring environment reproducibility across local development and cloud CI/CD by robustly handling dynamic internal package resolutions
- ▶ Implemented automated AWS S3 object tagging to optimize storage costs by automatically purging old files

SKILLS

Programming: Python · C++ · CUDA · HTML/CSS · $\text{\LaTeX}$

ML & Frameworks: PyTorch · PyTorch Lightning · Weights & Biases · einops · jaxtyping · beartype

DevOps & Tools: AWS S3 · Docker · Terraform · CI/CD (Bitbucket Pipelines) · pytest · mypy

Languages: Spanish (Native) · English (Fluent) · German (Basic)